CLASS 8 — MATHS · FORMULA SHEET

Chapter 11: **Exploring Some Geometric Themes** | Quick revision, one page

Polyhedra

- A polyhedron has faces (flat polygons), edges (where two faces meet) and vertices (corners).
- Euler's formula: $F + V - E = 2$ for convex polyhedra.
- Prism with n-sided base: $F = n + 2$, $V = 2n$, $E = 3n$.
- Pyramid with n-sided base: $F = n + 1$, $V = n + 1$, $E = 2n$.

Nets

- A net is a flat pattern that folds into a solid; a cube has 11 different nets of 6 squares.
- Number of faces of the solid $=$ number of polygons in its net.

Painted cube ($n \times n \times n$ cut into n^3 unit cubes, $n \geq 2$)

- 3 faces painted: 8 (corners). 2 faces painted: $12(n-2)$.
- 1 face painted: $6(n-2)^2$. No face painted: $(n-2)^3$.

Fractals

- A fractal is a pattern that repeats itself at smaller and smaller scales (self-similar).
- Sierpinski triangle: the number of triangles multiplies by 3 at each stage, giving 3^n at stage n.
- Koch snowflake: each stage replaces every side by 4 segments of one-third the length.

MASTERMIND